

Note 4.5 Indeterminate Forms and L'Hospital's Rule

Indeterminate Forms

- **Type $\frac{0}{0}$:**

For a limit of form $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ where $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$, then the limit is called an indeterminate form of type $\frac{0}{0}$.

- **Type $\frac{\infty}{\infty}$:**

For a limit of form $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ where $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = \infty$, then the limit is called an indeterminate form of type $\frac{\infty}{\infty}$.

- **L'Hospital's Rule:**

Suppose $f(x)$ and $g(x)$ are differentiable and $g'(x) \neq 0$ on a open interval that contains a (except possibly a). Suppose that

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$$

or that

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = \infty$$

Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

if the limit on the right side of the equation exists.

Indeterminate Forms $\infty \cdot 0$, $\infty - \infty$

- $\infty \cdot 0$:

$$f(x) \cdot g(x) = \frac{f(x)}{\frac{1}{g(x)}} \text{ or } \frac{g(x)}{\frac{1}{f(x)}}$$

- $\infty - \infty$:

Convert difference into a quotient so that we have a indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

Indeterminate Powers

- The limit

$$\lim_{x \rightarrow a} f(x)^{g(x)}$$

Rewrite the function as an exponential:

$$f(x)^{g(x)} = e^{g(x) \ln f(x)}$$

then $g(x) \ln f(x)$ is of type $0 \cdot \infty$